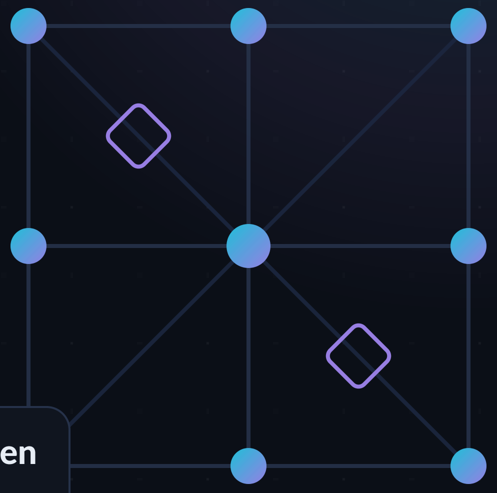
SEA QUANTATHON 2026 · DEEP-TECH QUANTUM COMPUTING

SYNDRA

The error-correction layer that makes quantum computers reliable enough to solve real problems



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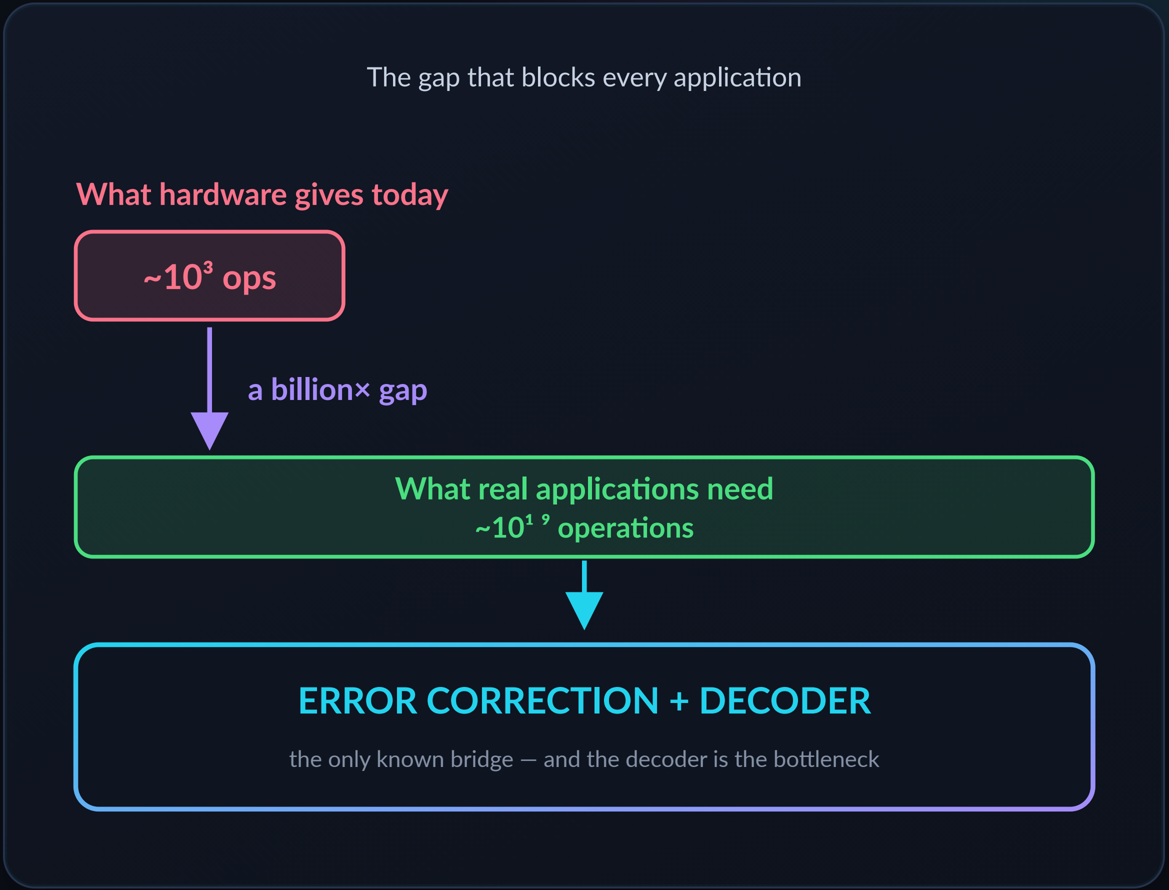
SDG 3 · 7 · 9 · 13

$16.3B market by 2030

Open-source · hardware-agnostic

Quantum computers are too error-prone to be useful

- Today's quantum chips make **an error roughly every 1,000 operations**. Real applications need **billions** of operations in a row.
- The fix is **error correction**: measure error signals every cycle, and a **decoder** works out what went wrong — in **microseconds**, or the computation is lost.
- **Urgency:** hardware is scaling **now** — $13B of public money committed, 100+ logical qubits announced in 2026. The decoder is what's holding the field back.
- **Without a working decoder, none of quantum's promised impact** — medicine, clean energy, climate — ever arrives.



No decoder, no bridge. **This is the blocker we work on.**

WHAT WE BUILD

A smarter proofreader for quantum computers

A quantum computer is like a page of text where letters randomly flip while you read. **Error correction is the proofreader** that spots and fixes them before you see the page. **We build that proofreader** — and we make it smarter using quantum machine learning.

THE PRODUCT

A software decoder

Not hardware. A **software library (SDK)** that plugs into a quantum computer's control system — the way a graphics driver plugs into a PC.

WHAT IT DOES

Reads alarms, fixes errors

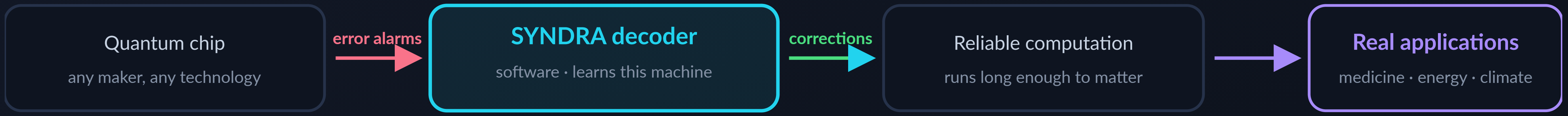
Every cycle the machine emits error signals. Our decoder reads them and outputs **what to correct** — continuously, while the program runs.

WHY OURS IS DIFFERENT

It learns the machine

Every quantum chip has its own noise fingerprint. Ours is **trained per machine**, so it adapts instead of assuming a textbook error model.

Where we sit: we don't build the machine — we make it trustworthy

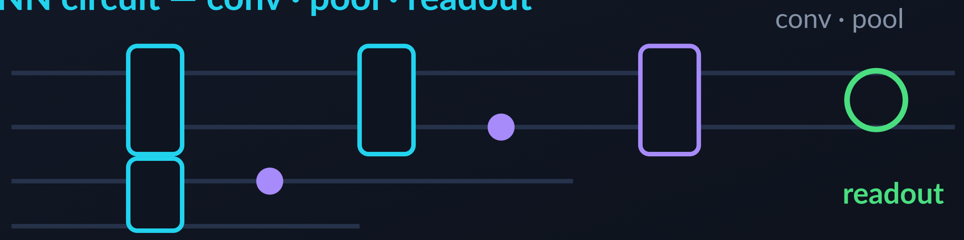


HOW IT WORKS

Quantum machine learning, used where it actually fits

- **Real QEC stack:** **Stim** + **PennyLane/PyTorch** vs **PyMatching** — the tools Google and Riverlane use.
- **QCNN for a reason:** Cong et al. proved it **barren-plateau-free** — gradients survive as qubits grow, where generic quantum ML cannot train.
- **1,102 parameters, flat in *d*:** **9× fewer** than our classical CNN, and **2.1× lower error** at d=3 on the same budget — only **35** are quantum.
- **No hand-built error model:** MWPM needs one, and reads only a binary syndrome. Learning fits leakage and analog readout that matching **cannot use at all.**

QCNN circuit — conv · pool · readout



MEASURED — REMOVE MEASURE QUBITS, D=3

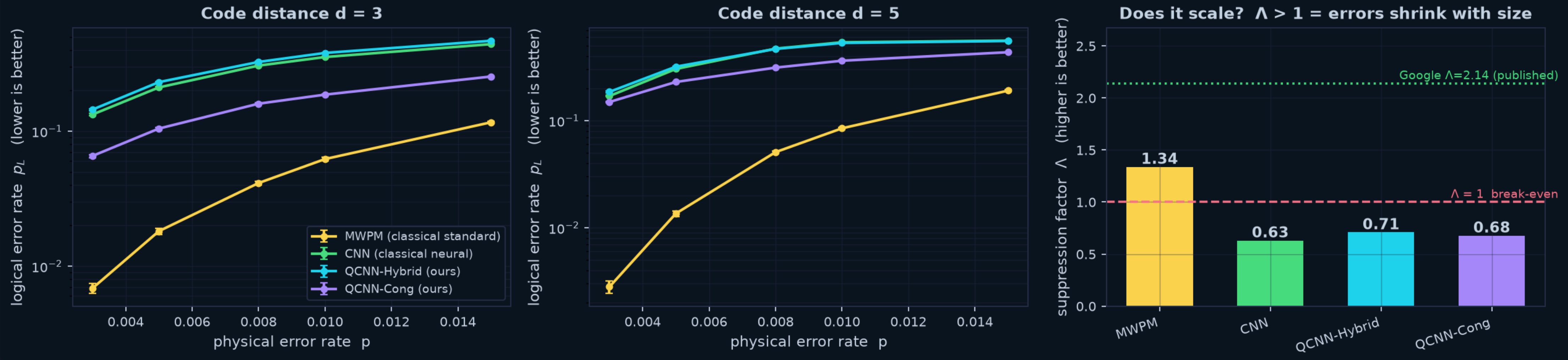
DETECTORS DROPPED	MWPM	OURS
none	0.042	0.141
12 of 24	0.157	0.159
degradation	3.78×	1.13×

Matching needs a **complete** syndrome. Learning does not — so a chip can lose measure qubits and still decode.

Ablation measured in this repo (d=3, p=0.008, 20k shots, identical shots for every decoder). Learned decoding without a hand-built noise model: Baireuther *et al.*, Quantum 2, 48 (2018); Lange *et al.*, PRR 7, 023181 (2025).

Among learned decoders, quantum wins

Measured on identical fresh test data — 20000 shots per point



Same data for all four

Identical fresh test set per point; test seed disjoint from training — **no leakage**.

Paper-exact metrics

Logical error rate, Λ suppression, error-per-cycle — from **Google, Nature 2024**.

Quantum beats classical ML

At **equal training budget**, our quantum decoder **Λ=0.71** vs classical neural **0.63**.

Honest finding

MWPM — a 40-year-old specialist algorithm — still leads overall. **We publish that.**

All points measured on fresh test data, 20k shots each. QCNN-Cong measured on an earlier architecture revision (2026-07-25); all others on the current build.

SDG 9 is our direct impact — the rest follows from it

SDG 9
DIRECT

WHAT WE ACTUALLY BUILD

Industry, Innovation & Infrastructure

Decoding is the layer that makes a quantum computer **usable at all**. We build that core infrastructure — open source, and reproducible on a laptop, so the capability is not locked inside a few proprietary stacks (**SDG 10**: access for emerging hubs like Southeast Asia).

INDIRECT — DOWNSTREAM, AND ONLY ONCE ERROR CORRECTION WORKS

SDG 3

Good Health & Well-being

Molecular simulation for drug discovery — projections cite up to **60% shorter** development cycles.

SDG 7

Affordable Clean Energy

Batteries, solar cells, catalysts — projected **30–40%** efficiency gains in renewables research.

SDG 13

Climate Action

Higher-fidelity climate and materials modelling — projected **40%** accuracy improvement.

Error correction works
SDG 9 — what we build

→

Quantum computers usable
long enough to solve real problems

→

SDG 3 · 7 · 13 outcomes become reachable
medicine · clean energy · climate

SDG impact figures are industry projections (UNDP Deep Tech Series; TCS; Pasqal; WEF 2024), not our measurements.

MARKET & BUSINESS MODEL

Who pays, how we ship, how we earn

MARKET BY 2030

$16.3B

From $2.2B in 2025, 33.7% CAGR. Every machine in it needs a decoder.

QEC FUNDING, 18 MONTHS

$400M

Raised by QEC-focused vendors alone — Riverlane, Q-CTRL, Quantum Machines.

PUBLIC COMMITMENT

$13B

US $2B, China $10B, EU €1B — QEC sits on every national roadmap.

CUSTOMERS

- **First customer: our challenge sponsor** — a photonic hardware maker whose stack we already run on.
- Then: other hardware makers, cloud QC platforms, national labs.

HOW WE SHIP

- Decoder SDK integrated into the partner's control stack.
- Open-source core for adoption; paid integration & tuning on top.

REVENUE

- Licensing by decode volume in production.
- Annual support + per-hardware tuning contracts.
- Co-development & national grant funding.

Sources: market.us, Precedence Research (market); TheQuantumInsider (QEC funding); NIST / Dept. of Commerce, EU Quantum Flagship (public funding).

COMPETITIVE LANDSCAPE

Where we sit — and the lane we take

DECODER	NEEDS A NOISE MODEL	LEARNS FROM HARDWARE DATA	PARAMS	OPENNESS
MWPM (classical standard)	yes	no — binary syndrome only	n/a	mature tooling
Google AlphaQubit	no	yes — leakage + soft readout	millions	proprietary
Riverlane Deltaflow (algorithmic)	yes	no	n/a	proprietary, $122M+
SYNDRA (ours)	no	yes, per machine	**1,102** — flat in *d*	open-source core

OUR WEDGE Only one decoder above **learns the machine's noise** — and it is closed, and **big**. On FPGA the binding constraint is DSP multiplier count, i.e. **parameters**, capping boards near $d = 13$. We take the open, **parameter-efficient** lane.

9×
FEWER PARAMS THAN OUR CLASSICAL CNN — AND 2.1× LOWER ERROR (D=3)

35
PARAMETERS IN THE QUANTUM CORE

100%
OPEN-SOURCE CORE

Params measured in this repo. FPGA DSP limit: Yang *et al.* arXiv:2605.04892. Riverlane = Collision Clustering / LCD, algorithmic; $122M+ total funding. AlphaQubit: Bausch *et al.*, Nature (2024).

FEASIBILITY — 1-YEAR INCUBATION

A concrete path from demo to deployed product

1 2 3 4

Months 1–3

Integrate

- Plug our SDK into the sponsor's control stack.
- Import their real device noise model.
- Reproduce their current decoder as the baseline to beat.

Months 4–6

Train & beat baseline

- Train on their noise fingerprint, not a textbook model.
- Target: cross $\Lambda > 1$, then beat MWPM.
- Deliver a reproducible benchmark report.

Months 7–9

Validate on hardware

- Run on the sponsor's real QPU via their cloud.
- Measure end-to-end decode latency in situ.
- Joint technical publication.

Months 10–12

Ship

- Production-grade SDK release, open-source core.
- Convert sponsor into **first paying customer**.
- Scale to further distances & second hardware partner.

DE-RISKED Months 1–3 are **already built**: working pipeline, 4 decoders, paper-exact metrics, GPU training, running today on the sponsor's cloud.

TEAM & THE ASK

Why this team, and what we need

Truong Le Gia Khanh
LEAD · QUANTUM RESEARCH
PhD candidate, University of Sydney. QEC theory, architecture design, scientific direction.

Chu Tuan Kiet
QUANTUM ALGORITHMS
University of Technology Sydney. QCNN circuit design, surface-code modelling.

Hoang Dinh Duy Anh
SOFTWARE ENGINEERING
FPT University. Pipeline, SDK architecture, reproducible test infrastructure.

Nguyen Gia Hien
ML & BENCHMARKING
FPT University. Model training, GPU optimisation, benchmark methodology.

4	50+	100%	12 mo
DECODERS BENCHMARKED	AUTOMATED TESTS, ALL PASSING	OPEN-SOURCE & REPRODUCIBLE	TO FIRST PAYING CUSTOMER

THE ASK Incubation support, compute credits, and seed funding — to turn a working benchmark into the **open error-correction layer** the quantum industry runs on. **Let's talk.**

Quantum computing will only matter if it becomes **reliable**.

We build the layer that makes it reliable.